

Aryan Varshney

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SUMMARY

AI/ML Engineer with hands-on experience building production Generative AI systems — real-time voice pipelines, RAG applications, and computer vision — using Python, LangChain, FAISS, and LLM APIs (LLaMA, Gemini). Strong foundations in DSA and end-to-end ML model development with Scikit-learn.

EXPERIENCE

EduLinkUp

Remote

AI/ML Intern — Generative AI (RAG, LLMs, FAISS, Prompt Engineering)

Jan 2026 – Feb 2026

- Built a Generative AI cybersecurity tool using Gemini 2.5 Flash with **RAG** over a **FAISS vector index** of **1,800+ document pages and 10 web sources**, enabling accurate, grounded responses to cybersecurity queries.

PROJECTS

Drishti : Swachh Bharat | 🐍 | Python, YOLOv8, Facenet512, FAISS, MediaPipe, SQLite

- Built a real-time AI littering surveillance pipeline that detects garbage drops via **YOLOv8**, crops the violator's face using **MediaPipe**, and dispatches an automated **e-challan (500 fine)** via SMTP — all without human intervention, achieving **zero false triggers** on legitimate dustbin disposal.
- Designed a **5-stage behavioral state machine** (Untracked → Attached → Detaching → Monitoring → Confirmed) that analyses how a person interacts with garbage over time, eliminating false alarms from people simply walking near trash or disposing waste in dustbins.
- Engineered **resolution-agnostic detection thresholds** scaled to each person's on-screen height, ensuring consistent accuracy across 720p webcams and 4K CCTV cameras, with **FAISS vector search** over facial embeddings for fast identity matching.

Dr. Chakshu | 🐍 | Python, PyTorch, DenseNet-121, Grad-CAM, FastAPI, FAISS, Groq Api

- Built a **multi-stage AI diagnostic pipeline** for chest X-rays using a fine-tuned DenseNet-121 backbone trained on **57K+ NIH CXR-14** images, achieving a mean **AUC-ROC of 0.7735** across Cardiomegaly, Pleural Effusion, and Pneumothorax with patient-aware splits to eliminate data leakage.
- Engineered an explainability layer via **Grad-CAM** exposed through a **FastAPI** endpoint that returns the highest-probability disease prediction alongside a Base64-encoded heatmap overlay, highlighting model attention regions on the denseblock4 convolutional layer.
- Designed a **RAG-augmented report generation** module using **FAISS** as a local vector store with sentence-transformers embeddings and the **Groq API** as the LLM, grounding AI-drafted structured radiology reports in retrieved medical-guideline context for clinical accuracy.

Vayu : Interactive Face Recognition Bot | 🐍 | Python, OpenCV, Facenet512, FAISS, SQLite, Sarvam AI

- Built a real-time AI booth assistant for IIIT Converge'26 that identified **50+ attendees** with **90% recognition accuracy** via FAISS cosine-similarity search over Facenet512 embeddings, delivering fully personalized spoken responses through a conversational pipeline, with average face recognition latency of **10ms per frame**.
- Developed an end-to-end voice pipeline using **Speech Recognition** (Speech-to-Text → LLaMA 3.3-70B → Text-to-Speech over WebSocket) with streaming audio playback that begins speaking within **0.4 seconds** of the first audio chunk arriving.
- Engineered concurrent execution using Python's **ProcessPoolExecutor** to run face recognition in an isolated process alongside live **Computer Vision (OpenCV)** and a **Silero VAD** audio thread, ensuring smooth, non-blocking operation.

EDUCATION

Jaypee Institute of Information Technology

Noida, India

B.Tech. in Computer Science and Engineering

2024 – 2028

TECHNICAL SKILLS

Programming Languages: Python, Java, Dart, C, C++

Frameworks: Django, FastAPI, Flask, Flutter, Spring Boot

Databases: PostgreSQL, MySQL, MongoDB

AI/ML: Machine Learning, Scikit-learn, PyTorch, HuggingFace, Embeddings, FAISS, NLP, LangChain, RAG

Tools: GitHub, Docker, REST APIs, WebSockets, n8n, Gemini API, Sarvam AI, Groq, Silero VAD

ACHIEVEMENTS

Keplay Certified

August 2025

- * **Top 168 of 18,000+ participants (top 1%)** in Keplay API Fellowship — selected through API design, testing, and documentation rounds